ConfAI 2026 · Poster Exhibition · Plaksha University

Recent Advances in AI · Responsible and Ethical AI track

Literature Discovery with Multi-Signal Ranking and Citation-Grounded RAG

Personalised arXiv recommendation · Selective full-text ingest · Grounded question answering

Ajay Sharma · M.S. in AI, Plaksha University · Supervisor: Dr. Subrat Panda · AI in Practice extended track

~3,830
full-text papers
in RAG index

~374k
section-aware
chunks embedded

~100
gold questions
in evaluation set

0.75
mean MRR
(curated index)

1 · PROBLEM

Finding papers is only half the work

Keyword search returns long, unranked lists that are not tailored to a research narrative. General chat models answer fluently but rarely with faithful links to source passages, so readers cannot tell support from invention. Researchers need a closed loop: discover papers for an interest, ingest a manageable full-text set, then ask questions and receive cited answers with honest limits when evidence is thin.

2 · SYSTEM LOOP

1. Discover

Multi-signal ranking of arXiv papers for free-text research interest (semantic + recency + citations + author continuity).

2. Select

Full text ingested only for chosen papers — PDF→Markdown (OCR fallback), section-aware chunks, local embeddings.

3. Ask

Dense retrieval over chunks; Gemini answers with paper-level citations and a coverage note when evidence is weak.

3 · METHOD HIGHLIGHTS

Ranking	Title/abstract embeddings fused with recency, citation impact and author continuity. Presets (Balanced, Semantic-heavy, Frontier) stay visible; editing switches to Custom.
Embeddings	EmbeddingGemma 2 (768-d) with optional Matryoshka truncation (512 / 256 / 128). Separate Chroma indexes for recommend vs full-text RAG.
Ingest	On-demand pipeline: download → pymupdf4llm Markdown → OCR when needed → chunk → embed. Resumable, memory-aware batching for laptop-scale runs.
Generation	Gemini for answers with citation instructions; failsafe path when the API is unavailable. Public demo uses MiniLM under free-tier memory limits.

Design choice
Only abstracts are indexed library-wide. Full text is ingested on demand so ranking stays cheap while RAG stays faithful to selected papers.

4 · EVALUATION

0.75
Mean MRR
95% CI ≈ 0.66–0.83

0.85
Recall@k
Labelled gold items

0.69
Hit@1
Top result correct

0.95
Cite validity
Valid citation indices

Protocol

~100 gold questions (definitions, methods, comparisons, multi-hop, unanswerable). Bootstrap CIs, latency percentiles, refusal metrics. Embedding dim sweep: 512-d matched 768-d MRR within ≈0.02.

Measured limits (reported openly)

Early metrics used a small curated full-text index; answers are weak when relevant papers were never ingested. Refusal precision lagged a 0.7 product bar. Broad keyword harvests can inject corpus noise.

5 · ENGINEERING

- CLI for batch ingest/debug; Streamlit for interactive demo
- Resumable ingest with progress files and hard per-paper timeouts
- Memory-aware micro-batching on laptop (MPS/CPU)
- Reproducible uv environments; one codebase for local and cloud

6 · TAKEAWAYS

- One loop from research interest → ranked papers → selective full text → cited answer.
- Ranking weights are inspectable; answers carry citations and coverage cues.
- Evaluation-first: gold set, CIs, and explicit failure modes — not only headline metrics.
- Next: refusal calibration, hybrid retrieval, cross-encoder re-rank, versioned gold gates.

Live demo
nataltiger26-arxiv-paper-suggester-rag-srcapp-olkakz.streamlit.app

Code
github.com/NatalTiger26/ArXiv_Paper_Suggester-RAG